The minimised expressions for D0, D1 and D2 are:

$$D_0 = \bar{Q}_0$$

$$D_1 = Q_1\bar{Q}_0 + \bar{Q}_1 Q_0$$

$$D_2 = Q_2\bar{Q}_0 + Q_2\bar{Q}_1 + Q_2Q_1Q_0 = Q_2(\bar{Q}_0 + \bar{Q}_1) + \bar{Q}_2Q_1Q_0 = Q_2(\overline{Q_0.Q_1}) + \bar{Q}_2(Q_1Q_0)$$

$$= Q_2 \oplus Q_1.Q_0$$

The complete circuit of the synchronous counter using positive edge triggered D FLIP-FLOPs is shown in figure below as

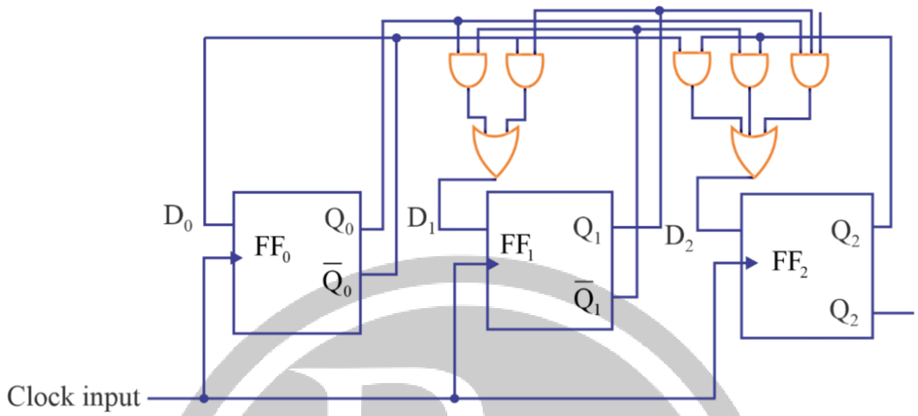

Fig. 4.40. 8-bit Synchronous Counter Circuit

4.12.2 Synchronous Sequential Circuit Models

A general block diagram of clocked sequential circuit is also known as finite stable machine (FSM). Depending upon the external outputs, there are two types of models of sequential circuits.

Mealy Model

In mealy model, the next of the function depends on present state as well as present inputs.

Moore Model

In more model, the next state depends on the present state. The block diagram of a Moore model is given as

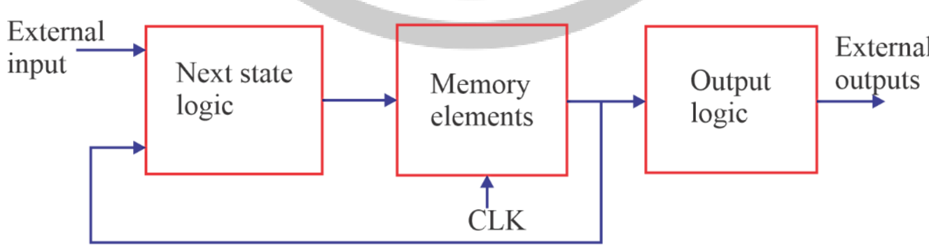

Fig. 4.41.

The systematic procedure for designing of clocked sequential circuit is based on the concept of 'state'. Hence the sequence of inputs, present & next states and output is represented by a state table or state diagram & if the procedure follows in the form of flow chart, it is known as algorithms state machine (ASM).

4.12.3. State Diagram

It is a directed graph, consisting of vertices (or nodes) and directed are between the nodes. Every state of the circuit is represented by a node in the graph. A node is represented by a circle with the name of the state written inside the circle. The directed are represents the state transitions.

With the circuit in may one state, at the occurrence of a clock pulse, there will be a state transition to the next state and there will be an output, corresponding to the requirement of the circuit. This state transition is represented by a directed line and we use each (/) for representing present state and the next state.

Example:

Draw the state diagram of D-flip-flop.

Solution:

The D flip-flop has only input (D) & two output states (Q = 0 & Q = 1).

Using the state table or characteristic table of the D-flip flop. The state diagram is given as

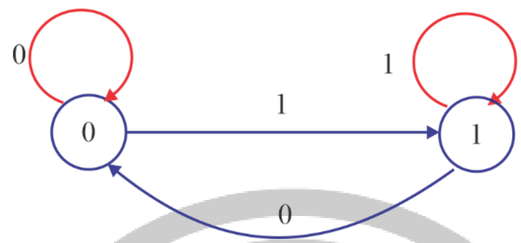

Fig. 4.42.

Example:

Draw the state diagram of a JK flip-flop.

Solution:

A JK flip-flop has inputs (J & K) and one clock input (CLK) and the two output states (Q = 0 & Q = 1).

Using the state table or characteristic table of the JK flip-flop, the state diagram is given as

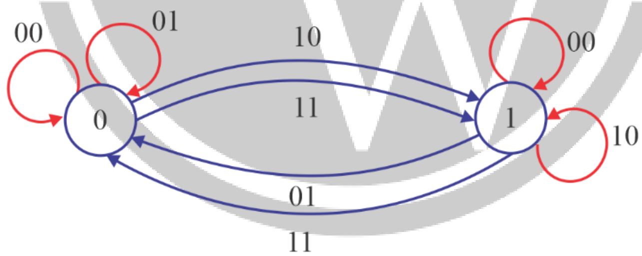

Fig. 4.43.